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EDUCATION

Madhav Institute of Technology and Science, Gwalior
B.Tech in Electronics Engineering

2023 – 2027
CGPA: 7.2

SKILLS

- **AI & LLM:** LangChain, RAG Pipelines, OpenAI API, Groq/Whisper, Anthropic Claude, HuggingFace, ChromaDB
- **Backend:** Python, FastAPI, RESTful APIs, SQL
- **ML & Deep Learning:** Supervised/Unsupervised Learning, Neural Networks, Transformers, Embeddings, Attention Mechanism
- **Libraries:** Pandas, NumPy, Matplotlib, Seaborn
- **Databases:** PostgreSQL, MySQL, ChromaDB, FAISS
- **Tools & Platforms:** Git, GitHub, Jupyter, VS Code, Cursor IDE

PROJECTS

AI Mock Interview Platform | React, FastAPI, Python, Monaco Editor, Groq API, Whisper, PostgreSQL, Neon
[Live Demo] | [GitHub]

- Built and deployed a full-stack AI interview platform supporting 3 interview modes (Coding, Technical, and Voice) with AI-generated questions, automated evaluation, and structured feedback reports.
- Engineered a modular FastAPI backend handling resume parsing, question generation, speech transcription, evaluation, and report generation through dedicated REST APIs backed by PostgreSQL.
- Developed a coding assessment environment using Monaco Editor with multi-language programming support, timer-based enforcement, and automatic submission locking to simulate real interview conditions.
- Integrated a multimodal voice interview pipeline using Groq Whisper (Speech-to-Text) and Web Speech API (Text-to-Speech) for real-time AI-driven interview interactions.
- Deployed the application on Render and Vercel, delivering a production-ready platform with persistent session management and GitHub-based version control.

AI Research Assistant | LangChain, FastAPI, Streamlit, ChromaDB, OpenAI | [GitHub]

- Built a Retrieval-Augmented Generation (RAG) system supporting multi-PDF document ingestion and cross-document question answering using semantic chunking, OpenAI embeddings, and ChromaDB vector search.
- Designed a modular FastAPI backend with dedicated endpoints for document ingestion and conversational retrieval, enabling the AI engine to function independently as a REST API.
- Enhanced retrieval quality through Max Marginal Relevance (MMR) search, reducing redundant context chunks and improving the diversity and relevance of retrieved information.
- Engineered retrieval workflows using LangChain LCEL components including RunnableParallel, RunnablePassthrough, prompt templates, and output parsers for scalable and maintainable AI pipelines.
- Developed an interactive Streamlit frontend with persistent chat history and session-state management, enabling seamless conversations across multiple uploaded documents.

EXPERIENCE

Summer Internship Program (Power BI)

Dec 2024

- Performed data cleaning and transformation for analysis.
- Built interactive dashboards using Power BI.
- Generated insights using charts, filters, and reports.

ACHIEVEMENTS

- Solved 100+ DSA Questions on LeetCode and GeeksforGeeks
- Certified in Generative AI, LLM and Responsible AI - Google Cloud Skills
- Earned Advanced SQL Badge and Python (Basic) Certificate from HackerRank.
- Finalist among 400+ teams in Frontend Battle 2K26 at MITS Gwalior.
- Completed a Generative AI Masterclass focused on prompt engineering and AI tools.